# LaunchLens

*LaunchLens helps startups and creators understand which marketing campaigns actually drive results.*
*"Turn marketing data into actionable insights."*

## The Problem

When people launch a product, they market it across multiple platforms: X, Reddit, LinkedIn, Product Hunt, Discord, Email, WhatsApp, and Influencers.

After a few days they know: *"I got 8,000 visitors."*

But that number alone isn't useful. Questions they still can't answer:

- Which campaign worked best?
- Which platform brought quality traffic?
- Which creator/influencer should I collaborate with again?
- Which campaign is wasting money?
- Which audience converts better?

That's the problem LaunchLens solves.

## Solution

LaunchLens creates **smart tracking links** for every campaign. Instead of sharing one website URL everywhere, users create separate links for every marketing campaign.

```
Twitter Launch    -> launchlens.app/r/TW123
Reddit Post       -> launchlens.app/r/RD221
LinkedIn          -> launchlens.app/r/LN882
Influencer        -> launchlens.app/r/INF45
```

Every click is tracked before redirecting to the real website. The visitor never notices.

## User Flow

```
Signup
  -> Create Project
  -> Create Campaign
  -> Generate Tracking Link
  -> Share Link
  -> Visitors Click
  -> LaunchLens Records Analytics
  -> Redirect to Actual Website
  -> Dashboard Updates
  -> AI Insights
```

## MVP Features

### Authentication

- JWT Authentication
- Login
- Signup

### Projects

```
Workspace -> Projects -> Campaigns
```

### Campaign Management

Example: Summer Launch — Product Hunt Launch, Twitter Thread, Reddit Post, LinkedIn Campaign.

### Smart Tracking Links

Short, campaign-specific URLs (e.g. `launchlens.app/r/abc123`) instead of long UTM URLs.

### Redirect Service

When a visitor clicks a tracking link, LaunchLens logs analytics, stores the data, and redirects instantly.

### Visitor Analytics

Every click stores: Timestamp, Browser, Device, OS, Referrer, Country, Campaign, and IP (hashed or anonymized).

### Dashboard

Shows Total Clicks, Unique Visitors, Campaign Performance, Top Countries, Browser Usage, Device Usage, and Traffic Timeline.

### Campaign Comparison

| Campaign | Clicks | Unique Visitors |
|---|---|---|
| Twitter | 1200 | 900 |
| Reddit | 500 | 450 |
| LinkedIn | 350 | 290 |

### AI Insights

Instead of showing only graphs, LaunchLens explains them:

> *Reddit generated fewer visitors but significantly higher engagement than Twitter.*

> *Most traffic came from mobile devices.*

> *Your LinkedIn campaign has a high percentage of repeat visitors.*

## Future Features

- Conversion Tracking
- Revenue Tracking
- ROI Dashboard
- Team Workspaces
- AI Recommendations
- A/B Testing
- Email Reports
- Custom Domains
- UTM Builder
- Influencer Analytics

## Tech Stack

| Frontend | Backend | Database / Other |
|---|---|---|
| React | Node.js | PostgreSQL |

| | | |
|---|---|---|
| TypeScript<br>Tailwind CSS | Express<br>JWT Auth | Prisma ORM<br>Recharts / Vercel / Railway |

## Database Tables

```
Users
Projects
Campaigns
TrackingLinks
ClickEvents
```

Simple and normalized.

## Architecture

```
React
  -> Express API
  -> PostgreSQL
  -> Analytics Queries
  -> Dashboard
  -> AI Summary
```

## Why This Project?

Because every founder asks:

> *"Which marketing actually works?"*

Google Analytics gives website analytics. LaunchLens focuses on **campaign-level marketing intelligence.**

## Interview Questions

**Why did you build LaunchLens?**

Because many founders promote products across multiple platforms but struggle to identify which campaigns actually perform well. Existing analytics tools provide traffic data, but campaign attribution is often fragmented or overly complex. I wanted to build a focused platform that makes campaign performance easy to understand.

**Why can't they just use Google Analytics?**

Google Analytics is excellent for website analytics, but LaunchLens is centered on **campaign management and attribution**. A user can create individual tracking links for each campaign and compare them in one place with AI-generated summaries.

**Why is visitor count useful?**

Visitor count alone isn't the goal — it becomes valuable when compared across campaigns. Example:

```
Twitter:   5000 visitors
LinkedIn:   900 visitors
```

At first glance, Twitter looks better. But suppose:

```
Twitter:   5000 visitors, 20 signups
LinkedIn:   900 visitors, 70 signups
```

Now LinkedIn is clearly the more effective channel. Even before adding conversion tracking, LaunchLens helps identify which campaigns drive meaningful engagement.

**What decisions can businesses make from visitor data?**

- Stop spending on poor-performing campaigns.
- Focus on the channels bringing quality traffic.
- Understand where their audience is located.
- Optimize for mobile if most visitors are on phones.
- Publish content when their audience is most active.
- Compare different marketing strategies objectively.

It's about making **better marketing decisions**, not just counting visitors.

**How is this different from a URL shortener?**

A URL shortener mainly creates short links. LaunchLens uses those links as the foundation for collecting campaign analytics and presenting actionable insights. The short URL is only one part of the system.

**What's technically challenging?**

- Designing a scalable database schema.
- Building a low-latency redirect service.
- Logging click events efficiently.
- Aggregating analytics for dashboards.
- Writing SQL queries for campaign comparisons.
- Presenting meaningful AI-generated summaries.

**How would you scale it?**

- Add indexes on frequently queried columns.
- Cache dashboard data using Redis.
- Queue analytics writes with BullMQ or Kafka.
- Use PostgreSQL partitioning for large click-event tables.
- Deploy multiple backend instances behind a load balancer.

**Why Node.js?**

Node.js is well suited because the application is primarily API-driven and I/O-bound (authentication, redirect handling, analytics logging, REST APIs, dashboard queries). It also keeps the frontend and backend in the same JavaScript/TypeScript ecosystem.

## Resume Description

> ***LaunchLens** — Built a full-stack marketing attribution platform using React, Node.js, Express, PostgreSQL, and Prisma. Developed campaign-specific tracking links, real-time analytics dashboards, AI-powered insight generation, JWT authentication, and interactive data visualizations to help startups evaluate marketing campaign performance.*

## What Makes This Stand Out

Most student projects are expense trackers, to-do apps, e-commerce clones, or chat apps.

LaunchLens solves a real problem faced by startups and marketers. It demonstrates:

- Backend API design
- Authentication
- Database modeling
- Analytics pipelines
- Data aggregation
- Dashboard development
- System design thinking
- Deployment

It gives you plenty to discuss in an SDE interview because it combines full-stack development with a practical business use case rather than being just another CRUD application.